PARUL CHEMICALS FSC

PC-SALES CRM & ERP PLATFORM

COMPLETE SYSTEM SPECIFICATION & ARCHITECTURAL MANUAL

Comprehensive Functional Guide Covering All 15 Modules, 32 Screens, 24 Database Schemas, System Workflows, and 246 API Specifications.

Document Title:	Complete System & Architecture Manual
Target Domain:	Dairy Cattle Nutrition & Agri-Vet Supply Chain ERP
Application Version:	v2.4.0 (Production Stable Release)
Target Audience:	Executives, Software Engineers, Operations Leads, System Admins
Frontend Host:	Vercel Edge Network (https://pc-sales.vercel.app)
Backend Host:	Render Cloud Infrastructure (Python 3.11 FastAPI)
Database:	Supabase PostgreSQL 15 Enterprise Cluster
Compilation Date:	October 31, 2026
Security Level:	CONFIDENTIAL — PROPRIETARY TO PARUL CHEMICALS

This document is automatically compiled from live codebase specifications and authoritative database schemas.

1. Executive Summary & System Architecture

1.1 Business Scope & Purpose

Parul Chemicals FSC operates across Gujarat, Maharashtra, and Madhya Pradesh manufacturing high-grade dairy cattle nutrition products, feed supplements, and mineral mixtures. The PC-Sales platform orchestrates customer relationship management, field sales routes, automated morning call distribution, regional multi-tier pricing, invoicing, payment debt recovery, Tally accounting synchronization, and HR recruitment.

1.2 Full-Stack Architecture

Layer	Technology	Key Responsibilities & Components
Frontend Client	React 18 + TypeScript 5 Vite + MUI v5	32 interactive screens, Material-UI design system, TanStack React Query cache, responsive mobile layout, universal voice dictation (Alt+V).
Backend API	FastAPI (Python 3.11) Render Hosting	246 REST endpoints, APScheduler 10:00 AM distribution cron, in-memory RBAC cache (10-min TTL), ReportLab publication engine, Pandas/OpenPyXL data cleaner.
Database Tier	Supabase PostgreSQL 15 PostgREST Engine	24 relational tables, B-tree indexes, foreign keys with ON DELETE CASCADE, JSONB survey documents, PostgreSQL sequence triggers.
Voice & AI	Sarvam AI Models & Web Speech API	Vernacular voice transcription in Gujarati, Hindi, and English. Integrated cattle survey questionnaire parser.
ERP Accounting	TallyPrime / Tally 9 XML Sync Pipeline	Bi-directional reconciliation of Sales, Receipts, and Journal vouchers against CRM orders with soft-delete purge.

2. Security Model & Role-Based Access Control (RBAC)

The system enforces a dual-layer security model: Client-side route gating via AuthContext and ProtectedRoute, paired with server-side endpoint protection via Depends(verify_permission('...')). Permissions are resolved in-memory with zero DB overhead on cache hits, refreshing every 10 minutes or instantly when roles are modified.

Role Key	Display Name	Operational Scope & Typical Assigned Permissions
admin	Admin	Full system access (all 52 permissions). Manages users, roles, pricing, logs, and financial records.
developer	Developer	Full system access equivalent to Admin for technical maintenance and operational audits.
sales_manager	Sales Manager	Approves orders, creates invoices, edits customers, manages payments, oversees distributors and shopkeepers.
logistics_manager	Logistics Manager	Manages order dispatch, shipment statuses, distributor accounts, and warehouse inventory fulfillment.
telecaller	Telecaller	Accesses daily calling list, executes customer calls, captures tri-lingual cattle survey, and books orders.
marketing_manager	Marketing Manager	Schedules product demos, monitors field officer trial conversions, runs call distribution.
business_analyst	Business Analyst	Read-only access across BI reports, analytics, financial trends, and executive dashboards.
product_manager	Product Manager	Controls product master, packaging capacities, regional price matrices, and catalog availability.
lead_manager	Lead Manager	Oversees entire CRM sales pipeline, monitors stage transitions, and assigns leads to representatives.
lead_owner	Lead Owner	Sales executive dedicated to qualifying prospects, executing field demos, and closing orders.
hr_&_administration	HR & Administration	Recruitment workspace, applicant screening, interview scorecards, job offers, and staffing logs.

3. Database Architecture & Data Dictionary

Below is the complete database dictionary for all core relational tables under Supabase PostgreSQL. Every column name, data type, integrity constraint, and business role is explicitly specified.

Table: customers — *PK: customer_id (SERIAL)*

Master directory of Sabhasad (dairy cattle farmers). Stores personal credentials, geographic location, Aadhaar number, status, and cattle profile survey.

Column	Type	Constraints / Default	Business Definition
customer_id	SERIAL	PRIMARY KEY	Unique auto-incrementing customer identifier
customer_code	TEXT	UNIQUE, NULLABLE	Unique business code for farmer (e.g. CUST001, 1002)
name	TEXT	NOT NULL	Full legal name of the farmer / Sabhasad
mobile	TEXT	INDEXED	Validated 10-digit Indian mobile number
village	TEXT	NULLABLE	Village name in uppercase
taluka	TEXT	NULLABLE	Sub-district / Taluka name in uppercase
district	TEXT	NULLABLE	District name in uppercase
state	TEXT	DEFAULT 'Gujarat'	Operating state (Gujarat, Maharashtra, MP)
adhar_no	TEXT	NULLABLE	12-digit Aadhaar identification number
status	TEXT	DEFAULT 'Active'	Account status ('Active', 'Inactive')
cattle_profile	JSONB	NULLABLE	Structured survey (milking count, dry cows, breed, feed type)
created_at	TIMESTAMP	CURRENT_TIMESTAMP	Record timestamp

Table: distributors — *PK: distributor_id (SERIAL)*

Regional stockists, wholesalers, and dairy cooperative union distribution centers.

Column	Type	Constraints / Default	Business Definition
distributor_id	SERIAL	PRIMARY KEY	Unique distributor identifier
name	TEXT	NOT NULL	Commercial distributor / business entity name
village	TEXT	NULLABLE	Operating village / town
taluka	TEXT	NULLABLE	Sub-district / Taluka
district	TEXT	NULLABLE	District location
mantri_name	TEXT	NULLABLE	Key executive / Mantri / Secretary contact
mantri_mobile	TEXT	NULLABLE	Mobile number of key contact
sabhasad_count	INTEGER	DEFAULT 0	Total affiliated dairy farmers
contact_in_group	INTEGER	DEFAULT 0	Group member count
status	TEXT	DEFAULT 'Active'	Account status ('Active', 'Inactive')
created_at	TIMESTAMP	CURRENT_TIMESTAMP	Creation timestamp

Table: shopkeepers — *PK: shopkeeper_id (SERIAL)*

Agro-retailers and village animal husbandry supply stores.

Column	Type	Constraints / Default	Business Definition
shopkeeper_id	SERIAL	PRIMARY KEY	Unique shopkeeper identifier
name	TEXT	NOT NULL	Shop / Owner name
village	TEXT	NULLABLE	Village location
taluka	TEXT	NULLABLE	Taluka / Tehsil
district	TEXT	NULLABLE	District location
mantri_name	TEXT	NULLABLE	Owner / Manager name
mantri_mobile	TEXT	NULLABLE	Contact phone number
sabhasad_count	INTEGER	DEFAULT 0	Total linked Sabhasads
sabhasad_morning	INTEGER	DEFAULT 0	Morning session participants
sabhasad_evening	INTEGER	DEFAULT 0	Evening session participants
milk_collection_morning	INTEGER	NULLABLE	Daily morning collection in Liters
milk_collection_evening	INTEGER	NULLABLE	Daily evening collection in Liters
nature_of_sabhasad	TEXT	NULLABLE	Farmer cooperative category
status	TEXT	DEFAULT 'Active'	Store operational status

Table: doctors — *PK: doctor_id (SERIAL)*

Veterinary practitioners and dairy health advisors.

Column	Type	Constraints / Default	Business Definition
doctor_id	SERIAL	PRIMARY KEY	Unique veterinary doctor identifier
name	TEXT	NOT NULL	Doctor / Practitioner name
village	TEXT	NULLABLE	Clinic village / headquarters
taluka	TEXT	NULLABLE	Sub-district
district	TEXT	NULLABLE	District
mantri_mobile	TEXT	NULLABLE	Direct mobile number
dairy_type	TEXT	NULLABLE	Cooperative / Private / Independent
support	TEXT	NULLABLE	Clinical support type
status	TEXT	DEFAULT 'Active'	Active practitioner status

Table: field_officers — *PK: field_officer_id (SERIAL)*

On-ground sales representatives, route officers, and trial demonstration executors.

Column	Type	Constraints / Default	Business Definition
field_officer_id	SERIAL	PRIMARY KEY	Unique field officer identifier
name	TEXT	NOT NULL	Full officer name
mobile	TEXT	NULLABLE	Phone number
assigned_districts	TEXT	NULLABLE	Comma-separated assigned districts
assigned_talukas	TEXT	NULLABLE	Assigned sub-districts
is_active	BOOLEAN	DEFAULT TRUE	Employment active status
created_at	TIMESTAMP	CURRENT_TIMESTAMP	Onboarding timestamp

Table: products — *PK: product_id (SERIAL)*

Master catalog of veterinary nutrition products, feed supplements, and mineral mixtures.

Column	Type	Constraints / Default	Business Definition
product_id	SERIAL	PRIMARY KEY	Unique product identifier
product_name	TEXT	NOT NULL	Commercial product name (e.g. Cal-Plus, Parul Mix)
packing_type	TEXT	NULLABLE	Packaging format (Bottle, Jar, Can, Bag, Pouch)
capacity_ltr	REAL	NULLABLE	Package volume / weight in Liters or Kilograms
category	TEXT	NULLABLE	Category (Mineral Mixture, Liquid Calcium, Bolus, Tonic)
standard_rate	REAL	NULLABLE	Base national catalog price
rate_gujarat	REAL	NULLABLE	Specific regional rate for Gujarat
rate_maharashtra	REAL	NULLABLE	Specific regional rate for Maharashtra
rate_mp	REAL	NULLABLE	Specific regional rate for Madhya Pradesh
custom_rates	JSONB	DEFAULT '{}'	Dynamic custom rate mapping by state/tier
is_active	INTEGER	DEFAULT 1	Catalog availability flag (1=Active, 0=Archived)

Table: sales — *PK: sale_id (SERIAL)*

Official sales orders, commercial invoices, and customer billing master records.

Column	Type	Constraints / Default	Business Definition
sale_id	SERIAL	PRIMARY KEY	Unique sale order identifier
invoice_no	TEXT	INDEXED	Unique generated tax invoice number (INV-YYYY-XXXX)
customer_id	INTEGER	FK -> customers	Customer / Sabhasad buyer
distributor_id	INTEGER	FK -> distributors	Optional distributor buyer
sale_date	TEXT	NOT NULL	ISO date string (YYYY-MM-DD)
total_amount	REAL	DEFAULT 0	Gross invoice monetary amount (INR)
total_liters	REAL	DEFAULT 0	Total volume / mass sold
payment_status	TEXT	DEFAULT 'Pending'	Status ('Paid', 'Partial', 'Pending')
shipment_status	TEXT	DEFAULT 'Pending'	Dispatch status ('Pending', 'Dispatched', 'Delivered')
notes	TEXT	NULLABLE	Order remarks / dispatch instructions
created_at	TIMESTAMP	CURRENT_TIMESTAMP	Order timestamp

Table: sale_items — *PK: sale_item_id (SERIAL)*

Line items breakdown for sales orders.

Column	Type	Constraints / Default	Business Definition
sale_item_id	SERIAL	PRIMARY KEY	Unique line item identifier
sale_id	INTEGER	FK -> sales	Parent sale order link
product_id	INTEGER	FK -> products	Linked product
quantity	INTEGER	NOT NULL	Number of units ordered
rate	REAL	NOT NULL	Unit billing rate applied
amount	REAL	NOT NULL	Total amount (quantity * rate)

Table: payments — *PK: payment_id (SERIAL)*

Cash collections, bank transfers, and invoice settlements.

Column	Type	Constraints / Default	Business Definition
payment_id	SERIAL	PRIMARY KEY	Unique transaction identifier
sale_id	INTEGER	FK -> sales	Settled invoice reference
payment_date	TEXT	NOT NULL	Date payment received
payment_method	TEXT	NULLABLE	Method ('Cash', 'RTGS/NEFT', 'UPI', 'Cheque')
amount	REAL	NOT NULL	Payment amount in INR
rrn	TEXT	NULLABLE	Bank Reference / UPI RRN / Cheque number
reference	TEXT	NULLABLE	Internal accounting reference
notes	TEXT	NULLABLE	Receipt notes

Table: demos — *PK: demo_id (SERIAL)*

On-field product trials, sample provisions, and conversion tracking.

Column	Type	Constraints / Default	Business Definition
demo_id	SERIAL	PRIMARY KEY	Unique demo record identifier
customer_id	INTEGER	FK -> customers	Target farmer / Sabhasad
distributor_id	INTEGER	FK -> distributors	Associated distributor
product_id	INTEGER	FK -> products	Demonstrated product
demo_date	TEXT	NOT NULL	Scheduled / executed demo date
quantity_provided	INTEGER	NULLABLE	Trial sample quantity given
follow_up_date	TEXT	NULLABLE	Scheduled farmer check-in date
conversion_status	TEXT	DEFAULT 'Scheduled'	Status ('Scheduled', 'Completed', 'Converted', 'Lost')
notes	TEXT	NULLABLE	Farmer trial feedback & response

Table: calling_assignments — *PK: assignment_id (SERIAL)*

Daily lead and farmer calling distribution queue for telecallers.

Column	Type	Constraints / Default	Business Definition
assignment_id	SERIAL	PRIMARY KEY	Queue assignment record ID
assigned_to	TEXT	NOT NULL	Telecaller user email address
customer_id	INTEGER	FK -> customers	Assigned customer target
distributor_id	INTEGER	NULLABLE	Assigned distributor target
entity_type	TEXT	DEFAULT 'customer'	Target type ('customer' or 'distributor')
priority_score	REAL	DEFAULT 0.0	Computed urgency/value score (0.0 to 100.0)
assigned_date	DATE	NOT NULL	Assignment date for daily roster
status	TEXT	DEFAULT 'pending'	Status ('pending', 'completed', 'rescheduled')

Table: call_logs — *PK: log_id (SERIAL)*

Detailed record of telephonic interactions, call outcomes, and cattle survey questionnaires.

Column	Type	Constraints / Default	Business Definition
log_id	SERIAL	PRIMARY KEY	Call record identifier
customer_id	INTEGER	NULLABLE	Customer recipient link
distributor_id	INTEGER	NULLABLE	Distributor recipient link
user_email	TEXT	NOT NULL	Telecaller email who made call
call_outcome	TEXT	NOT NULL	Outcome ('Interested', 'Order Placed', 'Ringing', 'Busy', 'Invalid')
notes	TEXT	NULLABLE	Conversation notes & customer remarks
called_at	TIMESTAMP	CURRENT_TIMESTAMP	Call timestamp
duration_seconds	INTEGER	DEFAULT 0	Call duration in seconds
survey_json	JSONB	NULLABLE	Tri-lingual cattle survey response JSON
next_follow_up	TIMESTAMP	NULLABLE	Agreed callback date/time

Table: telecaller_orders — *PK: order_id (SERIAL)*

Telephonically captured sales orders awaiting managerial approval.

Column	Type	Constraints / Default	Business Definition
order_id	SERIAL	PRIMARY KEY	Telecaller order identifier
customer_id	INTEGER	NOT NULL	Target customer ID
telecaller_email	TEXT	NOT NULL	Creator telecaller email
order_date	DATE	NOT NULL	Order booking date
total_amount	REAL	DEFAULT 0	Total order monetary value
status	TEXT	DEFAULT 'Pending'	Approval state ('Pending', 'Approved', 'Rejected')
notes	TEXT	NULLABLE	Booking notes
approved_by	TEXT	NULLABLE	Manager email who authorized
approved_at	TIMESTAMP	NULLABLE	Managerial approval timestamp
rejection_reason	TEXT	NULLABLE	Reason if rejected

Table: telecaller_order_items — *PK: id (SERIAL)*

Line items breakdown for telecaller-booked orders.

Column	Type	Constraints / Default	Business Definition
id	SERIAL	PRIMARY KEY	Line item ID
order_id	INTEGER	FK -> telecaller_orders	Parent telecaller order ID
product_id	INTEGER	FK -> products	Ordered product
quantity	INTEGER	NOT NULL	Quantity ordered
rate	REAL	NOT NULL	Unit price applied
amount	REAL	NOT NULL	Total line item cost

Table: telecaller_attendance — *PK: id (SERIAL)*

Morning attendance logs determining daily call distribution eligibility.

Column	Type	Constraints / Default	Business Definition
id	SERIAL	PRIMARY KEY	Attendance record ID
user_email	TEXT	NOT NULL	Telecaller email
attendance_date	DATE	NOT NULL	Shift date
is_present	BOOLEAN	DEFAULT FALSE	Presence confirmation flag
created_at	TIMESTAMP	CURRENT_TIMESTAMP	Log timestamp

Table: app_users — *PK: user_id (SERIAL)*

Authenticated system user credentials, employee identity, and role assignment.

Column	Type	Constraints / Default	Business Definition
user_id	SERIAL	PRIMARY KEY	User account identifier
email	TEXT	UNIQUE, NOT NULL	Login email address
name	TEXT	NOT NULL	Full employee name
role	TEXT	NOT NULL	Assigned RBAC role key (e.g. admin, sales_manager)
password_hash	TEXT	NOT NULL	Bcrypt hashed password
is_active	BOOLEAN	DEFAULT TRUE	Account active flag
created_at	TIMESTAMP	CURRENT_TIMESTAMP	User creation date

Table: roles — *PK: role_id (SERIAL)*

System and custom RBAC security roles with fast-path permission arrays.

Column	Type	Constraints / Default	Business Definition
role_id	SERIAL	PRIMARY KEY	Role record identifier
role_key	TEXT	UNIQUE, NOT NULL	Machine identifier (e.g. 'sales_manager')
display_name	TEXT	NOT NULL	Human friendly title
description	TEXT	NULLABLE	Scope and responsibilities
is_system	BOOLEAN	DEFAULT FALSE	Protected immutable system role flag
permission_keys	TEXT[]	DEFAULT '{}'	Fast-path array of assigned permission keys
created_at	TIMESTAMP	CURRENT_TIMESTAMP	Role creation timestamp

Table: permissions — *PK: permission_id (SERIAL)*

Granular capability keys gating backend endpoints and frontend features.

Column	Type	Constraints / Default	Business Definition
permission_id	SERIAL	PRIMARY KEY	Permission record identifier
permission_key	TEXT	UNIQUE, NOT NULL	String key (e.g. 'create_sale')
module	TEXT	NOT NULL	Categorization module
description	TEXT	NULLABLE	Action authorized by permission

Table: activity_logs — *PK: log_id (SERIAL)*

Comprehensive forensic audit trail recording field-level before/after diffs.

Column	Type	Constraints / Default	Business Definition
log_id	SERIAL	PRIMARY KEY	Audit log identifier
user_email	TEXT	NOT NULL	Actor employee email
action_type	TEXT	NOT NULL	Action ('CREATE', 'UPDATE', 'DELETE', 'LOGIN')
action_description	TEXT	NOT NULL	Human readable action summary
entity_type	TEXT	NULLABLE	Entity ('customer', 'sale', 'doctor', etc.)
entity_id	INTEGER	NULLABLE	Target entity primary key
entity_name	TEXT	NULLABLE	Target entity label
metadata	JSONB	NULLABLE	Detailed JSON payload including field diffs
ip_address	TEXT	NULLABLE	Client IP address
created_at	TIMESTAMP	CURRENT_TIMESTAMP	Audit timestamp

Table: tally_vouchers — *PK: id (SERIAL)*

Synchronized financial accounting vouchers from TallyPrime / Tally 9.

Column	Type	Constraints / Default	Business Definition
id	SERIAL	PRIMARY KEY	Internal voucher ID
guid	TEXT	UNIQUE	Tally global unique identifier
voucher_type	TEXT	INDEXED	Type ('Sales', 'Receipt', 'Journal', 'Contra')
voucher_number	TEXT	NULLABLE	Tally invoice / voucher number
date	DATE	NOT NULL	Voucher accounting date
amount	REAL	NOT NULL	Monetary transaction total
party_ledger_name	TEXT	NULLABLE	Account / Debtor / Creditor ledger
company_name	TEXT	NULLABLE	Tally company division
is_reconciled	BOOLEAN	DEFAULT FALSE	CRM matching reconciliation flag
synced_at	TIMESTAMP	CURRENT_TIMESTAMP	Ingestion timestamp

Table: recruitment_leads — *PK: candidate_id (SERIAL)*

Job candidates and applicants within the HR recruitment funnel.

Column	Type	Constraints / Default	Business Definition
candidate_id	SERIAL	PRIMARY KEY	Applicant identifier
name	TEXT	NOT NULL	Candidate full name
mobile	TEXT	NOT NULL	Candidate phone number
email	TEXT	NULLABLE	Email address
role_applied	TEXT	NOT NULL	Position (Field Officer, Telecaller, Manager)
experience_years	REAL	DEFAULT 0.0	Total professional experience
status	TEXT	DEFAULT 'Applied'	Hiring stage ('Applied', 'Screening', 'Interview', 'Offered', 'Hired', 'Rejected')
resume_url	TEXT	NULLABLE	Storage link to resume document
created_at	TIMESTAMP	CURRENT_TIMESTAMP	Application date

Table: user_sessions — *PK: session_id (SERIAL)*

Active user sessions and heartbeat tracking for concurrency and security audit.

Column	Type	Constraints / Default	Business Definition
session_id	SERIAL	PRIMARY KEY	Session identifier
user_email	TEXT	NOT NULL	User email address
login_time	TIMESTAMP	CURRENT_TIMESTAMP	Initial session establishment
last_heartbeat	TIMESTAMP	CURRENT_TIMESTAMP	Most recent 60-second ping
ip_address	TEXT	NULLABLE	Client IP address
user_agent	TEXT	NULLABLE	Browser and operating system signature
is_active	BOOLEAN	DEFAULT TRUE	Live session state

4. Module-by-Module Operational Specifications

MOD-01 — Authentication, RBAC & User Administration

Domain: Security & Administration | Tables: app_users, roles, permissions, role_permissions, activity_logs, user_sessions

Governs system access, multi-tenant RBAC permissions, audit trail logging, and live session heartbeats.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Login Screen	/login	Login.tsx	Public	Credential verification, JWT/Header issuance, and role extraction.
User Management	/user-management	UserManagement.tsx	view_users	Account creation, active toggling, role assignment, and password resets.
Role Management	/role-management	RoleManagement.tsx	manage_roles	RBAC matrix to create custom roles and assign from 52 granular permissions.
Admin Overview	/admin	Admin.tsx	view_activity_logs	High-level administrative system metrics, database health, and server stats.
Activity Logs	/activity	Activity.tsx	Authenticated	Forensic audit trail with field-level before/after diffs, user email, and IP.

Core Business Workflows:

- User logs in -> Credentials verified -> User permissions resolved and cached in memory (10-min TTL) and AuthContext.
- Client requests protected endpoint -> x-user-email checked -> Fast-path array checked in-memory without database hit.
- System mutations trigger ActivityLogger.log_update_with_diff() recording exact field changes into activity_logs.
- Browser sends 60-second heartbeat to maintain active session presence and audit concurrency.

MOD-02 — Sabhasad (Farmer / Customer) Management

Domain: Customer Relationship Management | Tables: customers, call_logs, sales, demos

Core registry of dairy cattle farmers. Handles phone sanitization, customer code deduplication, location tagging, and purchase summaries.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Sabhasad Directory	/customers	Customers.tsx	view_customers	Master directory of 10,000+ dairy farmers with 7-field search, phone duplicate detection, and full CRUD.

Core Business Workflows:

- Farmer Onboarding: Form input -> Frontend phone validation -> Duplicate probe -> Backend +91/0 prefix strip -> DB insert.
- Customer Summary: Aggregates lifetime sales, paid amount, pending dues, cattle survey JSON, and last 20 call logs.
- Fuzzy Search: Debounced PostgREST ilike search across Name, Mobile, Village, Taluka, District, State, and Code.

MOD-03 — Channel Partner Network (Distributors, Shopkeepers, Doctors, Field Officers)

Domain: Channel & Field Operations | Tables: distributors, shopkeepers, doctors, field_officers

Manages the multi-tier agricultural distribution channel from cooperative unions down to village dairy collection centers.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Distributors Directory	/distributors	Distributors.tsx	view_distributors	Wholesalers and union distributors with linked Sabhasad counts and Mantri contacts.
Shopkeepers Directory	/shopkeepers	Shopkeepers.tsx	view_shopkeepers	Village agro-retailers with morning/evening milk collection volumes and participation counts.
Veterinary Doctors	/doctors	Doctors.tsx	view_doctors	Practicing veterinary surgeons, prescription history, and clinic associations.
Field Officers	/field-officers	FieldOfficers.tsx	view_field_officers	On-ground sales officers, territory assignments (districts/talukas), and demo execution.

Core Business Workflows:

- Distributor Registration: Mantri details, affiliated Sabhasad counts, and location mappings registered.
- Shopkeeper Tracking: Daily milk intake (morning vs evening), payment recovery demo vs dispatch tracking.
- Field Officer Route Allocation: Specific districts and talukas bound to officers for on-ground sales and demos.

MOD-04 — Product Catalog & Dynamic Regional Pricing

Domain: Product & Inventory Management | Tables: products, product_regions

Maintains product formulas, packaging formats, and state-specific tiered pricing structures.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Product Pricing Matrix	/product-pricing	ProductPricing.tsx	manage_pricing	Product master catalog with packaging types, capacity, and regional price tiers (Gujarat, Maharashtra, MP, Custom).

Core Business Workflows:

- Product Setup: Standard rates defined for baseline packaging (Bottles, Cans, Pouches, Bags).
- Regional Pricing: Pricing managers configure distinct state rates or custom JSON rate overrides.
- Order Rate Lookup: Automatically resolves the matching unit rate during order creation based on customer state.

MOD-05 — Sales Order Booking, Invoicing & Dispatch

Domain: Sales & Order Fulfillment | Tables: sales, sale_items, telecaller_orders, telecaller_order_items

End-to-end sales lifecycle from telecaller booking and managerial validation to formal tax invoicing and dispatch.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Sales & Invoicing	/sales	Sales.tsx	view_sales	Direct order desk with multi-item grid, auto-calculated regional rates, and printable invoice PDF generation.
Order Management	/orders	OrderManagement.tsx	view_orders	Managerial approval desk for telecaller-booked orders. Allows review, approval, or rejection with reason.

Core Business Workflows:

- Telecaller Booking: Telecaller creates telecaller_order with line items -> enters 'Pending' queue.
- Managerial Approval: Sales Manager inspects order on OrderManagement -> approves order -> transforms into sales record.
- Direct Sale Booking: Staff creates direct sale -> Postgres trigger assigns next sequential invoice_no -> creates invoice PDF.

MOD-06 — Payment Collections & Financial Ledger

Domain: Finance & Collections | Tables: payments, sales

Handles customer debt recovery, partial payments, installment logging, and automated invoice payment status reconciliation.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Payments Desk	/payments	Payments.tsx	view_payments	Collection register tracking receipts, payment methods (Cash, UPI, RTGS, Cheque), RRNs, and invoice balance updates.

Core Business Workflows:

- Payment Receipt: Collector logs amount received, payment method, bank RRN reference, and linked invoice.
- Automatic Status Update: Backend recalculates total payments against invoice -> flips payment_status to 'Paid' or 'Partial'.
- Collection Reports: Generates exportable summaries for daily cash flow and debtor aging.

MOD-07 — Product Demonstrations & Field Officer Scheduling

Domain: Field Marketing & Demos | Tables: demos, field_officers, products, customers

Coordinates on-farm product trials of cattle feed supplements, farmer feedback collection, and lead conversion.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Product Demo Logs	/demos	Demos.tsx	view_demos	Registry of product trials, sample quantities provided, farmer reactions, and conversion outcomes.
Demo Scheduler	/demo-scheduler	DemoScheduler.tsx	view_demos	Calendar scheduler assigning field officers to farmers for demonstration visits with time slots.

Core Business Workflows:

- Trial Scheduling: Farmer requested trial -> Scheduler assigns field officer, product, and scheduled visit date/time.
- Execution & Sampling: Officer conducts demo, provides sample pack (e.g. 1 kg Mineral Mixture), logs observations.
- Trial Conversion: Follow-up call verifies results -> converts successful trial into regular monthly orders.

MOD-08 — Automated Call Distribution & Telecaller Duty Roster

Domain: Telecalling Operations | Tables: calling_assignments, telecaller_attendance, duty_sheet_log, call_logs

Intelligent, load-aware call distribution engine balancing customer calls among telecallers every morning.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Call Distribution	/call-distribution	CallDistribution.tsx	run_call_distribution	Workload balancer, 10 AM automatic assignment, telecaller capacity monitor, and priority scoring.
Telecaller Calling List	/calling-list	CallingList.tsx	view_calling_list	Telecaller terminal with click-to-call, customer profile, call outcome logging, and tri-lingual survey form.

Core Business Workflows:

- Morning Duty Sheet: Telecallers mark morning attendance -> Only present telecallers are eligible for assignments.
- 10:00 AM Cron Distribution: System evaluates unassigned customers -> computes priority score -> assigns balanced batches.
- Call Execution: Telecaller calls farmer -> selects outcome ('Interested', 'Order Placed', 'Ringing', etc.) -> records cattle survey.

MOD-09 — AI Voice-to-Text & Call Intelligence Engine

Domain: Artificial Intelligence & Voice | Tables: call_logs

Vernacular voice dictation engine integrating Web Speech API and Sarvam AI models for multilingual dairy farming terminology.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Voice Input Component	Global Component	VoiceTextField.tsx	Global	Universal microphone dictation supporting Gujarati, Hindi, and English with real-time waveform and Alt+V shortcut.

Core Business Workflows:

- Live Speech Recognition: User speaks -> Browser speech recognition streams interim transcripts -> merges with existing input.
- Sarvam AI Audio Fallback: MediaRecorder chunks sent to backend Sarvam AI service for high-accuracy vernacular transcription.
- Cattle Survey Ingestion: Voice responses parsed into structured cattle_profile JSON attached to customer records.

MOD-10 — Lead Pipeline & CRM Workspace

Domain: Lead Management & Sales Funnel | Tables: customers, call_logs, notes, demos

Comprehensive CRM sales funnel tracking prospects from initial inquiry to qualified Sabhasad buyer.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Lead Dashboard	/lead-dashboard	LeadDashboard.tsx	view_lead_dashboard	Lead funnel metrics, channel acquisition breakdown, conversion rates, and pipeline health.
Lead Kanban Pipeline	/leads	LeadPipeline.tsx	view_all_leads	Drag-and-drop visual pipeline across stages: New, Contacted, Qualified, Demo, Won, Lost.
Lead Detail Workspace	/lead-workspace	LeadWorkspace.tsx	work_leads	Individual lead detail cockpit with interaction timeline, follow-up scheduling, and quotation drafting.
Lead History	/lead-history	LeadHistory.tsx	work_leads	Full chronological audit log of all communications, calls, and status shifts for a lead.

Core Business Workflows:

- Lead Capture: Inbound inquiries from marketing or exhibitions recorded with source and dairy size.
- Stage Progression: Sales rep works lead through Kanban stages -> logs follow-up dates and task reminders.
- Conversion: Won leads automatically transition into active Sabhasad records with first order booking.

MOD-11 — Tally ERP Integration & Voucher Reconciliation

Domain: ERP Integration & Accounting | Tables: tally_vouchers, sales, payments

Bridges the operational PC-Sales CRM with corporate Tally ERP financial ledgers.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Tally Data Integration	/tally-data	TallyData.tsx	view_tally_data	Sync station for TallyPrime/Tally 9 XML & Excel vouchers. Reconciles Sales, Receipts, and Journal entries.

Core Business Workflows:

- Voucher Ingestion: Batch import of XML/Excel voucher dumps exported from Tally ERP.
- Cross-Reconciliation: Matches Tally Sales/Receipt vouchers with CRM invoices and payments by date, party, and amount.
- Audit & Cleanup: Soft-deletes superseded entries and identifies unmatched financial discrepancies.

MOD-12 — Excel Data Import & Ingestion Pipeline

Domain: Data Engineering & Migration | Tables: import_history, customers, distributors

Enterprise data ingestion engine transforming messy legacy spreadsheets into clean database records.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Data Import Hub	/import	DataImport.tsx	import_data	Bulk ingestion station with fuzzy alias matching, data validation preview, conflict detection, and batch insertion.

Core Business Workflows:

- Spreadsheet Upload: User uploads raw Excel (.xlsx, .xls) containing farmer or distributor lists.
- Preprocessing & Alias Resolution: Python backend maps aliases (e.g. 'Farmer Name' -> 'name'), parses phone numbers, infers morning/evening split columns.
- Preview Grid: Frontend highlights duplicates, validation warnings, and new rows for user confirmation.
- Batch Ingestion: One-click commit inserts validated rows and creates permanent audit entry in import_history.

MOD-13 — Executive BI Dashboard & Advanced Reporting

Domain: Business Intelligence & Analytics | Tables: sales, payments, customers, products

Real-time executive metrics, sales velocity tracking, and automated PDF/Excel business report generation.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Executive Dashboard	/dashboard	Dashboard.tsx	view_dashboard	Real-time command center showing gross revenue, active customers, pending collections, and top products.
BI Reports Hub	/reports	Reports.tsx	view_reports	Multi-dimensional reporting console generating custom PDF & Excel reports by district, taluka, product, and agent.

Core Business Workflows:

- Dashboard Aggregation: Real-time queries calculate monthly targets, collection percentages, and top customers.
- Report Generation: Users specify date ranges and dimensions -> backend compiles publication-grade PDF and multi-tab Excel workbooks.

MOD-14 — Predictive Demand Forecasting & Scoring Engine

Domain: Machine Learning & Algorithms | Tables: sales, customers, distributors

Empowers supply chain managers with predictive demand forecasting and telecallers with prioritized calling scores.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Demand Forecasting	/forecasting	Forecasting.tsx	run_forecasting	Statistical inventory prediction projecting product demand 30/60/90 days ahead based on seasonal patterns.
Scoring Algorithm	/algorithm	Algorithm.tsx	run_algorithm	ML-based scoring console rating farmers and distributors by conversion likelihood (0-100 score).

Core Business Workflows:

- Time-Series Forecasting: Evaluates 12-month historical sales -> projects required inventory by state and product.
- Customer Scoring: Scoring engine calculates weights for order frequency, monetary value, recency, and cattle herd size.

MOD-15 — HR Recruitment & Talent Acquisition System

Domain: Human Resources & Recruitment | Tables: recruitment_leads, recruitment_activities

Streamlines hiring of field officers, telecallers, and administrative staff from resume screening to employment offers.

Frontend User Interfaces:

Screen Name	Route	Component	Permission	Capabilities & UI Elements
Recruitment Workspace	/recruitment	RecruitmentWorkspace.tsx	view_recruitment	Complete ATS pipeline managing applicants, interview stages, evaluation scorecards, and hiring offers.

Core Business Workflows:

- Candidate Onboarding: HR enters applicant details, experience, role applied, and resume document link.
- Multi-stage Evaluation: Candidate progresses through Screening -> Round 1 -> Technical Interview -> HR Round.
- Offer & Onboarding: Evaluator records scorecard ratings -> system generates offer letter -> onboards into app_users.

5. End-to-End Operational Lifecycle Blueprints

Workflow	Trigger / Stage	Step-by-Step Technical & Business Sequence
Flow 1: Sabhasad Onboarding	Lead / Farmer Contact	Phone validation -> Duplicate probe -> Backend +91/0 cleanup -> customer_code uniqueness check -> Supabase insert -> ActivityLogger diff record.
Flow 2: 10:00 AM Call Distribution	Morning Shift Routine	Telecallers log presence -> 10 AM cron evaluates present telecallers -> Scores unassigned leads by priority -> Distributes evenly in calling_assignments queue.
Flow 3: Telecalling & Cattle Survey	Daily Farmer Calls	Telecaller opens /calling-list -> Dials farmer -> Uses Alt+V voice typing -> Captures tri-lingual survey (cow/buffalo count, milk yield) -> Records call outcome.
Flow 4: Telecaller Order Approval	Order Booking to Dispatch	Telecaller books order in terminal -> Enters Pending queue -> Sales Manager reviews in /orders -> Approves -> Transforms into official sale -> Warehouse dispatches.
Flow 5: Multi-Region Invoicing	Billing & Payment Recovery	Buyer selected -> State identified (Gujarat / Maharashtra / MP) -> Regional price matrix applied -> Invoice PDF generated -> Payment recorded with RRN -> Balance updated.
Flow 6: Tally ERP Reconciliation	Financial Synchronization	Accountant exports XML/Excel from Tally -> Uploads into /tally-data -> Vouchers mapped to CRM customers -> Cross-reconciliation verifies discrepancies -> Stale entries purged.

6. Master API Directory (All 246 Endpoints)

Complete authoritative catalog of all 246 API endpoints registered in the FastAPI server. Organized by HTTP Method, Path, Function Handler, and Functional Category.

#	Method	Route Path	Function Handler	Category
1	GET/HEAD	/openapi.json	openapi	Core
2	GET/HEAD	/docs	swagger_ui_html	Core
3	GET/HEAD	/docs/oauth2-redirect	swagger_ui_redirect	Core
4	GET/HEAD	/redoc	redoc_html	Core
5	GET	/	root	Core
6	GET	/health	health_check	Core
7	POST	/cron/distribute	cron_distribute	Core
8	POST	/cron/midnight-refresh	cron_midnight_refresh	Core
9	POST	/cron/nightly-scoring	cron_nightly_scoring	Core
10	GET	/api/customers/check-phone	check_customer_phone	Customers
11	GET	/api/customers/	get_customers	Customers
12	GET	/api/customers/search	search_customers	Customers
13	GET	/api/customers/{customer_id}/summary	get_customer_summary	Customers
14	GET	/api/customers/{customer_id}	get_customer	Customers
15	POST	/api/customers/	create_customer	Customers
16	PUT	/api/customers/{customer_id}	update_customer	Customers
17	DELETE	/api/customers/{customer_id}	delete_customer	Customers
18	GET	/api/products/	get_products	Products
19	GET	/api/products/all	get_all_products	Products
20	GET	/api/products/config/regions	get_product_regions	Products
21	POST	/api/products/config/regions	create_product_region	Products
22	GET	/api/products/config/categories	get_product_categories	Products
23	POST	/api/products/config/categories	create_product_category	Products
24	DELETE	/api/products/config/regions/{region_name}	delete_product_region	Products
25	DELETE	/api/products/config/categories/{category_name}	delete_product_category	Products
26	GET	/api/products/{product_id}	get_product	Products
27	POST	/api/products/	create_product	Products
28	PUT	/api/products/{product_id}	update_product	Products
29	DELETE	/api/products/{product_id}	delete_product	Products
30	GET	/api/sales/	get_sales	Sales
31	GET	/api/sales/pending-payments	sales_with_pending	Sales
32	GET	/api/sales/refund-due	sales_with_refund_due	Sales
33	POST	/api/sales/	create_sale	Sales
34	POST	/api/sales/confirm	confirm_sales	Sales
35	GET	/api/sales/{sale_id}	get_sale	Sales
36	PUT	/api/sales/{sale_id}	update_sale	Sales
37	DELETE	/api/sales/{sale_id}	delete_sale	Sales
38	GET	/api/sales/{sale_id}/invoice-pdf	get_invoice_pdf	Sales
39	GET	/api/payments/	get_payments	Payments
40	GET	/api/payments/pending	get_pending	Payments
41	GET	/api/payments/sale/{sale_id}	get_payment_history	Payments

42	**GET**	/api/payments/{payment_id}	get_payment	Payments
43	**POST**	/api/payments/test	test_payment_creation	Payments
44	**POST**	/api/payments/	create_payment	Payments
45	**PUT**	/api/payments/{payment_id}	update_payment	Payments
46	**DELETE**	/api/payments/{payment_id}	delete_payment	Payments
47	**GET**	/api/notes/	list_notes	Core
48	**GET**	/api/notes/sale/{sale_id}	get_notes_for_sale	Core
49	**GET**	/api/notes/sale/{sale_id}/items	get_sale_items_for_return	Core
50	**POST**	/api/notes/	create_note	Core
51	**PATCH**	/api/notes/{note_id}/void	void_note	Core
52	**PATCH**	/api/notes/{note_id}/pickup_status	update_pickup_status	Core
53	**GET**	/api/demos/suggestions	get_demo_suggestions	Demos
54	**GET**	/api/demos/redemo	get_redemo_history	Demos
55	**GET**	/api/demos/	get_demos	Demos
56	**GET**	/api/demos/{demo_id}	get_demo	Demos
57	**POST**	/api/demos/	create_demo	Demos
58	**PUT**	/api/demos/{demo_id}	update_demo	Demos
59	**PUT**	/api/demos/{demo_id}/status	update_demo_status	Demos
60	**DELETE**	/api/demos/{demo_id}	delete_demo	Demos
61	**GET**	/api/distributors/check-phone	check_distributor_phone	Distributors
62	**GET**	/api/distributors/search	search_distributors	Distributors
63	**GET**	/api/distributors/resolved	get_resolved_distributors	Distributors
64	**GET**	/api/distributors/	get_distributors	Distributors
65	**GET**	/api/distributors/calculator	get_distributors_for_calculator	Distributors
66	**POST**	/api/distributors/	create_distributor	Distributors
67	**PUT**	/api/distributors/{distributor_id}	update_distributor	Distributors
68	**DELETE**	/api/distributors/{distributor_id}	delete_distributor	Distributors
69	**GET**	/api/shopkeepers/	get_shopkeepers	Shopkeepers
70	**POST**	/api/shopkeepers/	create_shopkeeper	Shopkeepers
71	**PUT**	/api/shopkeepers/{shopkeeper_id}	update_shopkeeper	Shopkeepers
72	**DELETE**	/api/shopkeepers/{shopkeeper_id}	delete_shopkeeper	Shopkeepers
73	**GET**	/api/doctors/	get_doctors	Doctors
74	**POST**	/api/doctors/	create_doctor	Doctors
75	**PUT**	/api/doctors/{doctor_id}	update_doctor	Doctors
76	**DELETE**	/api/doctors/{doctor_id}	delete_doctor	Doctors
77	**GET**	/api/field-officers/	get_field_officers	Field Officers
78	**POST**	/api/field-officers/	create_field_officer	Field Officers
79	**PUT**	/api/field-officers/{field_officer_id}	update_field_officer	Field Officers
80	**DELETE**	/api/field-officers/{field_officer_id}	delete_field_officer	Field Officers
81	**GET**	/api/dashboard/metrics	dashboard_metrics	Dashboard
82	**GET**	/api/dashboard/collected-payments	get_collected_payments	Dashboard
83	**GET**	/api/dashboard/sales-trend	sales_trend	Dashboard
84	**GET**	/api/dashboard/recent-sales	recent_sales	Dashboard
85	**GET**	/api/dashboard/upcoming-demos	upcoming_demos	Dashboard
86	**GET**	/api/dashboard/system-start-date	system_start_date	Dashboard
87	**GET**	/api/reports/sales-trend	get_sales_trend	BI Reports
88	**GET**	/api/reports/analytics-summary	get_analytics_summary	BI Reports
89	**GET**	/api/reports/dimension-breakdown	get_dimension_breakdown	BI Reports
90	**GET**	/api/reports/filter-options	get_filter_options	BI Reports
91	**GET**	/api/reports/payment-trend	get_payment_trend	BI Reports

92	**GET**	/api/reports/sales-order-summary-pdf	generate_sales_order_summary_pdf	BI Reports
93	**GET**	/api/reports/sales-summary	sales_summary	BI Reports
94	**GET**	/api/reports/customers-pdf	generate_customers_pdf	BI Reports
95	**GET**	/api/reports/invoices-pdf	generate_invoices_pdf	BI Reports
96	**GET**	/api/reports/payments-pdf	generate_payments_pdf	BI Reports
97	**GET**	/api/reports/calling-list-pdf	generate_calling_list_pdf	BI Reports
98	**GET**	/api/reports/sales-analytics-pdf	get_sales_analytics_pdf	BI Reports
99	**GET**	/api/reports/sales-analytics-excel	get_sales_analytics_excel	BI Reports
100	**GET**	/api/reports/product-report-pdf	get_product_report_pdf	BI Reports
101	**GET**	/api/reports/product-report-excel	get_product_report_excel	BI Reports
102	**GET**	/api/reports/customer-analytics-pdf	get_customer_analytics_pdf	BI Reports
103	**GET**	/api/reports/customer-analytics-excel	get_customer_analytics_excel	BI Reports
104	**GET**	/api/reports/telecaller-dashboard	get_telecaller_dashboard_route	BI Reports
105	**GET**	/api/reports/telecaller/{report_type}/{format}	download_telecaller_report	BI Reports
106	**GET**	/api/reports/telecaller/charts	get_telecaller_charts_api	BI Reports
107	**GET**	/api/analytics/payment-distribution	payment_distribution	Core
108	**GET**	/api/admin/my-logs	get_my_activity_logs	Admin & Users
109	**GET**	/api/admin/health	health	Admin & Users
110	**GET**	/api/admin/check-status	check_user_status	Admin & Users
111	**GET**	/api/admin/activity-logs	get_activity_logs	Admin & Users
112	**GET**	/api/admin/activity-logs/stats	get_activity_stats	Admin & Users
113	**GET**	/api/admin/users	get_all_users	Admin & Users
114	**GET**	/api/admin/app-users	get_app_users	Admin & Users
115	**DELETE**	/api/admin/activity-logs/{log_id}	delete_activity_log	Admin & Users
116	**DELETE**	/api/admin/activity-logs/bulk	delete_old_activity_logs	Admin & Users
117	**PUT**	/api/admin/update-product-price/{product_id}	update_product_price	Admin & Users
118	**POST**	/api/admin/update-product-prices-bulk	update_product_prices_bulk	Admin & Users
119	**POST**	/api/admin/users	create_user	Admin & Users
120	**PATCH**	/api/admin/users/{target_email}/status	set_user_status	Admin & Users
121	**PATCH**	/api/admin/users/{target_email}/profile	update_user_profile	Admin & Users
122	**PATCH**	/api/admin/users/{target_email}/password	reset_user_password	Admin & Users
123	**PATCH**	/api/admin/users/{target_email}/role	update_user_role	Admin & Users
124	**GET**	/api/admin/user-sessions	get_all_user_sessions	Admin & Users
125	**GET**	/api/admin/user-sessions/history	get_user_session_history	Admin & Users
126	**POST**	/api/algorithm/run	run_algorithm	Algorithms
127	**POST**	/api/imports/distributors/confirm-import	confirm_import_distributors	Data Import
128	**POST**	/api/imports/distributors/preprocess	preprocess_distributors	Data Import
129	**POST**	/api/imports/sabhasad/preprocess	preprocess_sabhasads	Data Import
130	**POST**	/api/imports/sabhasad/confirm-import	confirm_import_sabhasads	Data Import
131	**POST**	/api/imports/excel	import_excel	Data Import
132	**GET**	/api/imports/history	get_import_history	Data Import
133	**DELETE**	/api/imports/history/{import_id}	delete_import_history	Data Import
134	**POST**	/api/imports/rollback-latest	rollback_latest_import	Data Import
135	**GET**	/api/automation/debug-state	debug_state	Call Automation
136	**GET**	/api/automation/my-assignments	get_my_assignments	Call Automation
137	**POST**	/api/automation/reorder-assignments	reorder_assignments	Call Automation
138	**GET**	/api/automation/my-callbacks	get_my_callbacks	Call Automation
139	**GET**	/api/automation/calling-summary	get_calling_summary	Call Automation
140	**POST**	/api/automation/start-call-timer	start_call_timer	Call Automation
141	**POST**	/api/automation/log-adhoc-call	log_adhoc_call	Call Automation

142	**POST**	/api/automation/update-call-status	update_call_status	Call Automation
143	**GET**	/api/automation/telecallers	get_telecallers	Call Automation
144	**GET**	/api/automation/admin/telecaller-profile	get_telecaller_profile	Call Automation
145	**GET**	/api/automation/admin/assignments	get_admin_assignments	Call Automation
146	**GET**	/api/automation/admin/export-assignments-excel	export_admin_assignments_excel	Call Automation
147	**POST**	/api/automation/admin/distribute-mantris	admin_distribute	Call Automation
148	**POST**	/api/automation/admin/distribute-sabhsads	distribute_sabhsads	Call Automation
149	**POST**	/api/automation/admin/undo-village-assignment	undo_village_assignment	Call Automation
150	**GET**	/api/automation/locations	get_locations	Call Automation
151	**POST**	/api/automation/admin/reassign	admin_reassign	Call Automation
152	**GET**	/api/automation/distribution-status	get_distribution_status	Call Automation
153	**POST**	/api/automation/admin/refresh-distribution	admin_refresh_distribution	Call Automation
154	**GET**	/api/automation/admin/available-counts	get_available_counts	Call Automation
155	**POST**	/api/automation/admin/bulk-reassign	admin_bulk_reassign	Call Automation
156	**POST**	/api/automation/admin/transfer-pending	admin_transfer_pending	Call Automation
157	**POST**	/api/automation/trigger-daily	trigger_daily_cron	Call Automation
158	**POST**	/api/automation/trigger-midnight	trigger_midnight_cron	Call Automation
159	**POST**	/api/automation/trigger-scoring	trigger_scoring_cron	Call Automation
160	**GET**	/api/notifications/	get_notifications	Notifications
161	**GET**	/api/notifications/unread-count	get_unread_count	Notifications
162	**GET**	/api/notifications/{notification_id}	get_notification	Notifications
163	**POST**	/api/notifications/	create_notification	Notifications
164	**PUT**	/api/notifications/mark-all-read	mark_all_read	Notifications
165	**PUT**	/api/notifications/{notification_id}/mark-read	mark_notification_read	Notifications
166	**DELETE**	/api/notifications/{notification_id}	delete_notification	Notifications
167	**DELETE**	/api/notifications/delete-old	delete_old_notifications	Notifications
168	**GET**	/api/rbac/my-permissions	get_my_permissions	RBAC Security
169	**GET**	/api/rbac/roles	get_all_roles	RBAC Security
170	**POST**	/api/rbac/roles	create_role	RBAC Security
171	**DELETE**	/api/rbac/roles/{role_id}	delete_role	RBAC Security
172	**GET**	/api/rbac/permissions	get_all_permissions	RBAC Security
173	**GET**	/api/rbac/roles/{role_id}/permissions	get_role_permission_ids	RBAC Security
174	**PUT**	/api/rbac/roles/{role_id}/permissions	update_role_permissions	RBAC Security
175	**POST**	/api/user-sessions/heartbeat	session_heartbeat	User Sessions
176	**GET**	/api/user-sessions/today	get_today_session	User Sessions
177	**POST**	/api/user-sessions/log-auth	log_auth_event	User Sessions
178	**GET**	/api/forecasting/monthly	get_monthly_forecast	Forecasting
179	**GET**	/api/forecasting/district	get_district_forecast	Forecasting
180	**GET**	/api/forecasting/village	get_village_forecast	Forecasting
181	**PUT**	/api/chat/messages/{message_id}	edit_message	Chat
182	**DELETE**	/api/chat/messages/{message_id}	delete_message	Chat
183	**GET**	/api/chat/messages/search	search_messages	Chat
184	**POST**	/api/leads/intake	intake_lead	CRM Leads
185	**POST**	/api/leads/intake/rfq	intake_rfq	CRM Leads
186	**GET**	/api/leads/	get_all_leads	CRM Leads
187	**GET**	/api/leads/stats/pipeline	get_pipeline_stats	CRM Leads
188	**GET**	/api/leads/my	get_my_leads	CRM Leads
189	**GET**	/api/leads/sources	get_lead_sources	CRM Leads
190	**POST**	/api/leads/sources	save_lead_source	CRM Leads
191	**DELETE**	/api/leads/sources/{source_id}	delete_lead_source	CRM Leads

192	**GET**	/api/leads/{lead_id}/activities	get_lead_activities	CRM Leads
193	**GET**	/api/leads/{lead_id}	get_lead	CRM Leads
194	**GET**	/api/leads/{lead_id}/detail	get_lead_detail_owner	CRM Leads
195	**POST**	/api/leads/{lead_id}/assign	assign_lead	CRM Leads
196	**POST**	/api/leads/{lead_id}/comment	manager_comment	CRM Leads
197	**DELETE**	/api/leads/{lead_id}	delete_lead	CRM Leads
198	**PATCH**	/api/leads/{lead_id}	update_lead	CRM Leads
199	**POST**	/api/leads/{lead_id}/activities	log_activity	CRM Leads
200	**POST**	/api/leads/{lead_id}/close	close_lead	CRM Leads
201	**GET**	/api/leads/owners/list	get_lead_owners	CRM Leads
202	**GET**	/api/leads/{lead_id}/quotation	get_quotation	CRM Leads
203	**GET**	/api/leads/{lead_id}/quotations/history	get_quotation_history	CRM Leads
204	**PUT**	/api/leads/{lead_id}/quotation	upsert_quotation	CRM Leads
205	**POST**	/api/leads/{lead_id}/quotation/commit	commit_quotation	CRM Leads
206	**POST**	/api/leads/{lead_id}/quotation/attachment	upload_quotation_attachment	CRM Leads
207	**DELETE**	/api/leads/{lead_id}/quotation/attachment	delete_quotation_attachment	CRM Leads
208	**GET**	/api/leads/{lead_id}/quotation/html	get_quotation_html	CRM Leads
209	**GET**	/api/attendance/duty-sheet-status	get_duty_sheet_status	Attendance
210	**GET**	/api/attendance/telecallers	get_all_telecallers	Attendance
211	**POST**	/api/attendance/submit-duty-sheet	submit_duty_sheet	Attendance
212	**GET**	/api/attendance/status	get_attendance_status	Attendance
213	**POST**	/api/attendance/mark	mark_attendance	Attendance
214	**GET**	/api/reviews/	get_all_reviews	Core
215	**DELETE**	/api/reviews/{review_id}	delete_review	Core
216	**POST**	/api/telecaller-orders/	create_telecaller_order	Telecaller Orders
217	**GET**	/api/telecaller-orders/	get_telecaller_orders	Telecaller Orders
218	**POST**	/api/telecaller-orders/{order_id}/telecaller-confirm	telecaller_confirm_order	Telecaller Orders
219	**GET**	/api/telecaller-orders/pending	get_pending_telecaller_orders	Telecaller Orders
220	**POST**	/api/telecaller-orders/bulk-mark-approved	bulk_mark_approved	Telecaller Orders
221	**POST**	/api/telecaller-orders/{order_id}/approve	approve_telecaller_order	Telecaller Orders
222	**POST**	/api/telecaller-orders/{order_id}/reject	reject_telecaller_order	Telecaller Orders
223	**GET**	/api/telecaller-orders/my-confirmation-calls	get_my_confirmation_calls	Telecaller Orders
224	**GET**	/api/telecaller-orders/export-merged-excel/{sale_id}	export_merged_excel	Telecaller Orders
225	**GET**	/api/telecaller-orders/export-confirmations-excel	export_confirmations_excel	Telecaller Orders
226	**POST**	/api/telecaller-orders/export-selected	export_selected_orders	Telecaller Orders
227	**GET**	/api/telecaller-orders/{order_id}	get_telecaller_order	Telecaller Orders
228	**POST**	/api/call-logger/transcribe-chunk	transcribe_chunk	Voice AI
229	**POST**	/api/call-logger/analyze	analyze_transcript	Voice AI
230	**GET**	/api/recruitment/categories	get_recruitment_categories	HR Recruitment
231	**GET**	/api/recruitment/leads/check-duplicate	check_duplicate_candidate	HR Recruitment
232	**GET**	/api/recruitment/leads/follow-ups	get_recruitment_follow_ups	HR Recruitment
233	**GET**	/api/recruitment/leads	get_recruitment_leads	HR Recruitment
234	**GET**	/api/recruitment/leads/{lead_id}	get_recruitment_lead	HR Recruitment
235	**GET**	/api/recruitment/leads/{lead_id}/activities	get_candidate_activities	HR Recruitment
236	**POST**	/api/recruitment/leads/{lead_id}/activities	record_candidate_activity	HR Recruitment
237	**POST**	/api/recruitment/leads/{lead_id}/status	change_candidate_status	HR Recruitment
238	**POST**	/api/recruitment/leads	create_recruitment_lead	HR Recruitment
239	**PATCH**	/api/recruitment/leads/{lead_id}	update_recruitment_lead	HR Recruitment
240	**GET**	/api/tally/ping	ping_backend	Tally ERP
241	**GET**	/api/tally/audit	audit_vouchers	Tally ERP

242	**POST**	/api/tally/ingest	ingest_vouchers	Tally ERP
243	**POST**	/api/tally/reconcile/cleanup	reconcile_cleanup	Tally ERP
244	**GET**	/api/tally/vouchers	list_vouchers	Tally ERP
245	**GET**	/api/tally/stats	get_stats	Tally ERP
246	**GET**	/api/tally/filter-options	get_filter_options	Tally ERP